

– Understanding LSTM – a tutorial into Long Short-Term Memory Recurrent Neural Networks

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Abstract

Long Short-Term Memory Recurrent Neural Networks (LSTM-RNN) are one of the most powerful dynamic classifiers publicly known. The network itself and the related learning algorithms are reasonably well documented to get an idea how it works. This paper will shed more light into understanding how LSTM-RNNs evolved and why they work impressively well, focusing on the early, ground-breaking publications. We significantly improved documentation and fixed a number of errors and inconsistencies that accumulated in previous publications. To support understanding we as well revised and unified the notation used.

1 Introduction

This article is an tutorial-like introduction initially developed as supplementary material for lectures focused on Artificial Intelligence. The interested reader can deepen his/her knowledge by understanding Long Short-Term Memory Recurrent Neural Networks (LSTM-RNN) considering its evolution since the early nineties. Today's publications on LSTM-RNN use a slightly different notation and a much more summarized representation of the derivations. Nevertheless the authors found the presented approach very helpful and we are confident this publication will find its audience.

Machine learning is concerned with the development of algorithms that automatically improve by practice. Ideally, the more the learning algorithm is run, the better the algorithm becomes. It is the task of the learning algorithm to create a classifier function from the training data presented. The performance of this built classifier is then measured by applying it to previously unseen data.

Artificial Neural Networks (ANN) are inspired by biological learning systems and loosely model their basic functions. Biological learning systems are